

# Arash Sarshar

Department of Computer Engineering and Computer Science  
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## EDUCATION

Ph.D. Computer Science, Virginia Tech, 2020  
M.S. Electrical Engineering, Ferdowsi University, 2012  
B.S. Electrical Engineering, Shahid Beheshti University, 2008

## APPOINTMENTS

2023–2030 California State University, Long Beach  
Assistant Professor, Department of Computer Engineering and Computer Science  
2020–2022 Virginia Tech  
Postdoctoral Associate, Department of Computer Science

## RESEARCH AREAS

Scientific Machine Learning: Data-Driven and Equations-Based Modeling, Simulation, and Optimization  
Computational Science: Numerical Methods for Forward and Inverse Problems, High-Performance Computing

## PUBLICATIONS

### Journal Articles

- 2025 A Raj, CE Gudumotou, S Bun, K Srinivasa, A Sarshar, “Deep Operator Networks for Bayesian Parameter Estimation in PDEs,” *Communications on Applied Mathematics and Computation*, .  
<https://doi.org/10.1016/j.cpc.2025.109853>
- 2025 J Barry-Straume, A Sarshar, AA Popov, A Sandu, “Physics-informed neural networks for PDE-constrained optimization and control,” *Computer Physics Communications*, 317.  
<https://doi.org/10.1007/s42967-025-00499-x>
- 2024 A. Bhattacharjee, A. Popov, A. Sarshar, A. Sandu, “Improving the Adaptive Moment Estimation (ADAM) stochastic optimizer through an Implicit-Explicit (IMEX) time-stepping approach,” *Journal of Machine Learning for Modeling and Computing*, 2024.  
<https://doi.org/10.1615/JMachLearnModelComput.2024053508>
- 2022 S. Roberts, A. Popov, A. Sarshar, A. Sandu, “A fast time-stepping strategy for ODE systems equipped with a surrogate model,” *SIAM Journal on Scientific Computing*, 44 (3), A1405–A1427.  
<https://epubs.siam.org/doi/abs/10.1137/20M1386281>
- 2022 S. González–Pinto, D. Hernández–Abreu, M. S. Pérez–Rodríguez, A. Sarshar, S. Roberts, A. Sandu, “A unified formulation of splitting-based implicit time integration schemes,” *Journal of Computational Physics*, Volume 448, 2022. <https://doi.org/10.1016/j.jcp.2021.110766>
- 2021 S. Roberts, J. Loeffeld, A. Sarshar, C. S. Woodward, A. Sandu, “Implicit multirate GARK methods,” *Journal of Scientific Computing*, Volume 87, Issue 1, 2021. <https://doi.org/10.1007/s10915-020-01400-z>

- 2021 A. Sarshar, S. Roberts, A. Sandu, “Alternating Directions Implicit Integration in a General Linear Method Framework,” *Journal of Computational and Applied Mathematics*, Volume 389, 15 January 2021. <https://doi.org/10.1016/j.cam.2019.112619>
- 2020 S. Roberts, A. Sarshar, A. Sandu, “Coupled Multirate Infinitesimal GARK Schemes for Stiff Systems with Multiple Time Scales,” *SIAM Journal on Scientific Computing*, Vol. 42, No. 3, PP A1609–A1638. <https://epubs.siam.org/doi/abs/10.1137/19M1266952>
- 2020 S. Roberts, A. Sarshar, A. Sandu, “Parallel Implicit-Explicit General Linear Methods,” *Communications in Applied Mathematics and Computational Science*, Volume 6, Issue 2, 2021. <https://doi.org/10.1007/s42967-020-00083-5>
- 2019 A. Sarshar, S. Roberts, A. Sandu, “Design of high-order decoupled multirate GARK schemes,” *SIAM Journal on Scientific Computing*, Vol. 41, No. 2, PP A816–A847. <https://epubs.siam.org/doi/abs/10.1137/18M1182875>
- 2017 A. Sarshar, P. Tranquilli, B. Pickering, A. McCall, A. Sandu, C. J. Roy, “A Numerical Investigation of Matrix-Free Implicit Time-Stepping Methods for Large CFD Simulations,” *Computers & Fluids*, Volume 159, 15 December 2017, Pages 53–63. <https://doi.org/10.1016/j.compfluid.2017.09.014>
- 2017 P. Tranquilli, S. R. Glandon, A. Sarshar, A. Sandu, “Analytical Jacobian-vector products for the matrix-free time integration of partial differential equations,” *Journal of Computational and Applied Mathematics*, Volume 310, 15 January 2017, Pages 213–223. <http://www.sciencedirect.com/science/article/pii/S0377042716302199>

### Conference Proceedings

- 2016 A. Sarshar, A. E. Tabari, “Design of a Stacked Stub-Loaded Patch Element for X-band Reflectarray Antenna with True Time Delay,” *7th European Conference on Antennas and Propagation*, Gothenberg, Sweden. <http://ieeexplore.ieee.org/xpl/abstractAuthors.jsp?reload=true&tp=&arnumber=6546931>

### Technical Reports

- 2022 A. A. Popov, A. Sarshar, A. Chenault, A. Sandu, “A Meta-learning Formulation of the Autoencoder Problem for Non-linear Dimensionality Reduction,” *Technical Report*, arXiv. <https://arxiv.org/pdf/2207.06676>
- 2019 A. Sarshar, S. Roberts, A. Sandu, “Linearly-Implicit General Linear Methods,” *Technical Report*, Virginia Tech. <https://vtechworks.lib.vt.edu/bitstream/handle/10919/109996/Technical-Report-Linearly-Implicit-GLM.pdf?sequence=1>

### INVITED TALKS

- 2024 “Uncertainty Quantification for Scientific Machine Learning Using Deep Operator Networks.” SIAM Annual Meeting 2024, Spokane, Washington. <https://youtu.be/iiTHm9u9lro>
- 2023 “Science-guided Machine Learning for Forward, Inverse, and Control Problems.” Grand Valley State University, Michigan. <https://people.cs.vt.edu/sarshar/talks/gvsu/>
- 2022 “On Time-stepping Methods for Gradient-flow Optimization.” Canadian Applied and Industrial Mathematics Society, SCMAI 2022, Ontario, Canada. <https://sarshar.dev/files/adam-gark-slides.pdf>
- 2020 “Linearly Implicit Genral Linear methods.” SIAM Conference on Computational Science and Engineering 2021, Atlanta, Georgia.
- 2019 “Discrete Multirate GARK schems.” SIAM Conference on Computational Science and Engineering 2019, Spokane, Washington. [https://www.pathlms.com/siam/courses/10878/sections/14361/video\\_presentations/127463](https://www.pathlms.com/siam/courses/10878/sections/14361/video_presentations/127463)

## CONFERENCE ACTIVITY

### Sessions Organized

- 2024 “New Methods in Probabilistic and Science-Guided Machine Learning.” SIAM Annual Meeting 2024, Spokane, Washington.
- 2021 “Advances in Time-Stepping Methods for Multiphysics and Multiscale Problems - Parts I & II.” SIAM Conference on Computational Science and Engineering, Atlanta, Georgia.

## GRANTS AND AWARDS

- 2024 CSULB ORED Multidisciplinary Research Grant (joint), “A Machine Learning-Based Multi-Scale, Multi-Fidelity Framework for Advanced Prediction of Thermodynamic Properties in Dense Gases.”

## COURSES TAUGHT

### California State University, Long Beach

Artificial Intelligence, Pattern Recognition, Machine Learning, Advanced Artificial Intelligence, Decision Making Under Uncertainty

### Virginia Tech

Numerical Methods

### Bahar Institute of Higher Education

Antennas and Wave Propagation

## SERVICE

### Service to the University

- 2025-2025 CSULB CECS RTP Revision Committee
- 2024-2025 CSULB CECS/STATS Data Science Degree Committee
- 2024-2026 CSULB COE Awards Committee
- 2024-2025 CSULB CECS Graduate Curriculum Committee
- 2023-2024 CSULB CECS Undergraduate Curriculum Committee

### Thesis Advising

- Carol Gudumotu, Msc. in Computer Science, Fall 2023
- Amogh Raj, Msc. in Computer Science, Fall 2024
- Keerthana Srinivasa Naidu, Msc. in Computer Science, Fall 2024

## PEER REVIEW

### Academic Journal Peer Review

- SIAM Journal on Scientific Computing*
- International Journal of Computer Mathematics*

*Computing and Visualization in Science*

*Geo-scientific Model Development*

*International Journal for Numerical Methods in Engineering*

*Journal of Applied Numerical Mathematics*

*Journal of Computational and Applied Mathematics*

*International Journal of Applied Mechanics*

*ICML Workshop on the Synergy of Scientific and Machine Learning Modeling*

## **MEMBERSHIPS**

Society for Industrial and Applied Mathematics

Institute of Electrical and Electronics Engineers

Association for Computing Machinery

## **PROFESSIONAL DEVELOPMENT**

2025 Beach Mentor Program Certificate, California State University, Long Beach.

2018 Graduate Certificate in Future Professoriate, Virginia Tech.

Updated October 2025